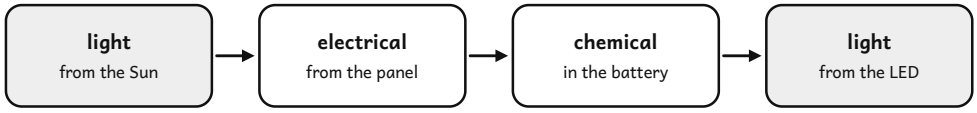
PHYSICAL AND CHEMICAL SCIENCE

Energy and Its Transformations

Forms of energy, energy chains, conservation of energy, heat and temperature



In this pack

Learn it: Energy: it moves and changes but is never lost ★☆☆
Fact check: Forms of energy ★☆☆
Think it through: Following the energy ★★☆
Investigate: Which wrapping keeps tea hottest? ★★☆
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This pack belongs to: _____ **Class:** _____

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LEARN IT · C4 · ENERGY

Energy: it moves and changes but is never lost

Name: Class: Date:

Learning goals

- Name different forms of energy and describe energy transfers.
- Explain that energy cannot be created or destroyed.
- Explain how temperature changes when an object gains or loses heat.

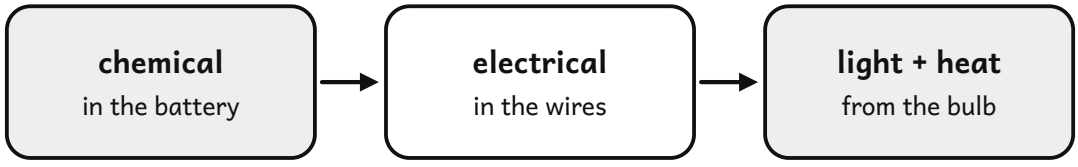
A rickshaw puller pedals through traffic. A ceiling fan spins. Rice cooks on a gas stove. A phone lights up. In every case, **energy** is moving from one place to another and changing from one form to another. Energy is what makes things happen.

Forms of energy

Form	Examples
kinetic (movement)	a moving bus, flowing river, spinning fan
potential (stored)	water held behind a dam, a stretched rubber band, a book on a high shelf
chemical (stored)	food, fuel (gas, petrol, wood), batteries
thermal (heat)	a hot cup of cha, the Sun's warmth
light	the Sun, a bulb, a firefly
sound	a dhol, a car horn, your voice
electrical	current flowing in wires

Energy transfers and chains

Energy is **transferred** from one thing to another, and **transformed** (changed) from one form to another. We can show this with an **energy chain**:



A torch.

- A rickshaw puller: chemical energy (food) → kinetic energy (moving rickshaw) + heat.
- A solar panel: light energy → electrical energy.
- A gas stove: chemical energy (gas) → thermal energy (heat) + light.

Energy is conserved

Energy **cannot be created or destroyed**. It only moves from place to place or changes form. So where does energy "go" when a fan is switched off? It has spread out into the surroundings, mostly as **heat** and **sound**. We often call this energy **wasted**, because it is not useful to us. An **efficient** machine wastes less energy. An LED bulb gives the same light as an old filament bulb but uses only about one fifth of the electricity, because it wastes much less as heat.

Heat and temperature

Temperature tells us how hot or cold something is, measured in °C with a **thermometer**. **Heat** is energy that moves from a **hotter** object to a **colder** one. When an object **gains** heat, its temperature **rises**; when it **loses** heat, its temperature **falls**. A hot cup of tea cools down until it reaches the temperature of the room.

Materials that let heat pass easily are **thermal conductors** (metals). Materials that slow heat down are **thermal insulators** (wood, plastic, wool, air). Cooking pots are metal, but their handles are wood or plastic.

Energy drives cycles

Energy from the Sun drives the **water cycle** (it evaporates water) and powers **food chains** (plants use light to make food). As energy flows through, **matter** (water, nutrients) goes round and round.

Bangladesh connection

Millions of homes in rural Bangladesh use **solar home systems**: a solar panel on the roof charges a battery that powers lights, fans and phone chargers at night.

Key words

energy	what makes things happen
kinetic energy	energy of movement
potential energy	stored energy
transfer	energy moving from one place or object to another
transform	energy changing from one form to another
efficient	wasting little energy
thermal conductor	a material that lets heat pass through easily
thermal insulator	a material that slows down heat moving through it

FACT CHECK · C4 · ENERGY

Forms of energy

★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Identify forms of energy.
- Complete energy chains.

1 Sort the examples by the main form of energy. [6]

SORT THESE

a spinning fan **a moving bus** **a plate of rice** **a drum beat** **a battery** **a firefly**

Kinetic	Chemical (stored)	Light or sound

2 Complete the energy chains. [8]

Device	Energy in	Useful energy out	Wasted energy
electric fan			
solar panel			
radio			
gas stove			

3 Fill in the gaps. [5]

WORD BANK

created **hotter** **colder** **conductors** **insulators** **falls** **rises**

Energy cannot be _____ or destroyed. Heat moves from _____ objects to _____ ones. When an object gains heat its temperature _____. Metals are good thermal _____.

Following the energy

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Explain energy transfers.
- Explain heating and cooling.

1 **Describe the energy changes when a rickshaw puller rides up a bridge and then freewheels down.** [3]

2 **"When a ball stops bouncing, its energy has been destroyed." Do you agree? Explain.** [2]

3 **Explain why a metal spoon in hot tea gets hot quickly, but a wooden spoon does not.** [2]

4 **A cold bottle of water and a hot cup of tea are left on a table. What happens to their temperatures? Explain using the word *heat*.** [2]

5 **Why is an LED bulb more efficient than a filament bulb?** [1]

INVESTIGATE · C4 · ENERGY

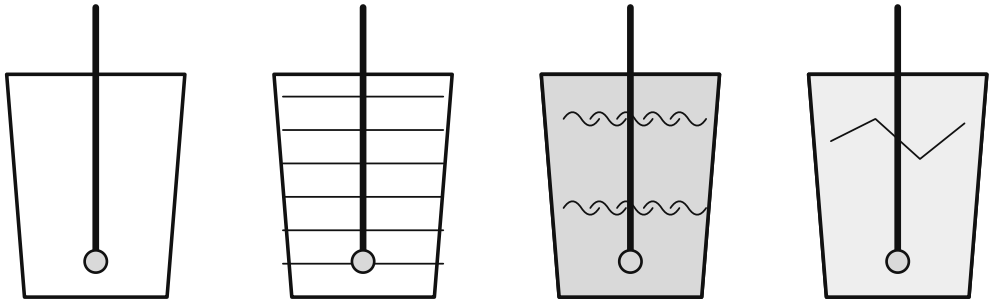
Which wrapping keeps tea hottest?

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Plan a fair test.
- Plot several lines on a graph.
- Explain results using insulation.



no cover **newspaper** **wool** **foil**

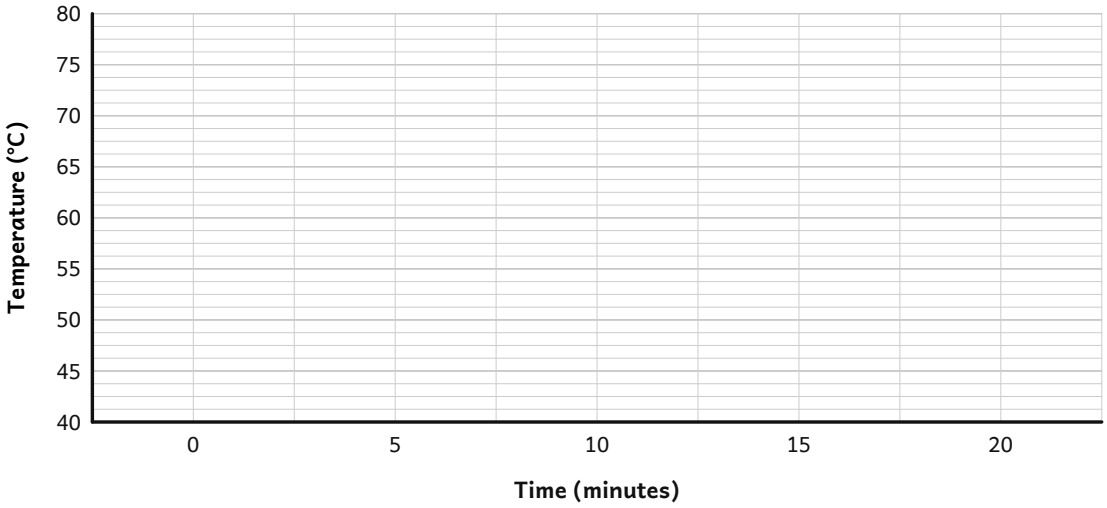
Class 5D poured 200 ml of water at 80 °C into four identical cups, wrapped three of them in different materials, and measured the temperature every 5 minutes.

1 **Name the variables.** [3]

Change	
Measure	
Keep the same	

2 **Here are the results. Plot all four on one line graph (use a different line or symbol for each).** [4]

Time (min)	0	5	10	15	20
no cover	80	72	64	58	52
newspaper	80	75	70	65	60
wool	80	77	74	71	68
foil	80	74	68	63	58



3 **Which material was the best insulator? Use numbers in your answer.** [2]

.....

.....

4 **Why do you think wool works so well?** [1]

.....

.....

5 **Suggest a real use for what the class found out.** [1]

.....

.....

REAL-WORLD CASE STUDY · C4 · ENERGY

Power from the sun: solar home systems

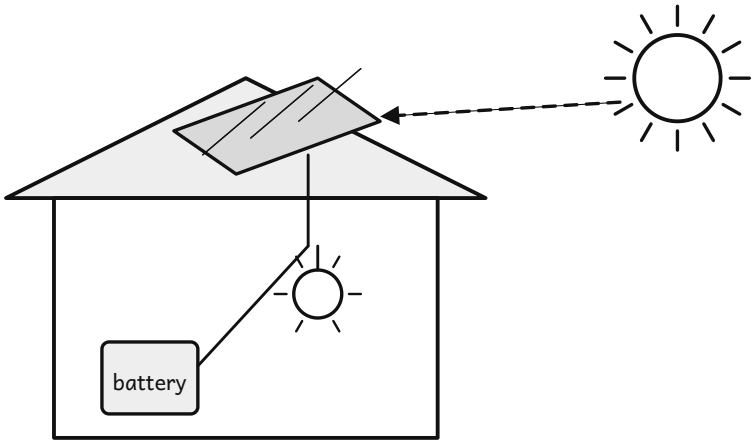
★★★ Grade 5

Name: **Class:** **Date:**

Learning goals

- Trace energy transfers in a real system.
- Use maths to plan energy use.
- Weigh benefits and problems.

Science · Maths · Economics · Environment · Technology



solar panel → battery → LED light, fan, phone charger

Twenty years ago, millions of village homes in Bangladesh had no electricity. At night families used **kerosene lamps**, which give dim light, smoky air and a risk of fire. Since the early 2000s, millions of homes have bought **solar home systems**, making Bangladesh one of the world's biggest users of small solar systems.

A solar panel on the roof turns sunlight into electricity during the day. This charges a **battery**. At night the battery powers **LED lights**, a small **fan**, a **phone charger** and sometimes a TV. Children can study after dark, and shops can stay open longer.

Problems include the **cost** of buying a system, batteries that wear out after a few years and must be **recycled** safely (old batteries contain harmful lead or chemicals), and less power on cloudy monsoon days.

Based on reports about Bangladesh's solar home system programme (IDCOL, World Bank).

1 **Write the energy chain for an LED light powered by a solar home system at night.** [3]

.....

.....

2 **Energy maths. A 50 W panel gets 5 hours of good sunshine a day.** [4]

a) How many watt-hours (Wh) of energy can it make in a day? (W × hours)

b) A family uses 3 LED bulbs of 5 W for 4 hours. How much energy is that?

c) They also use a 20 W fan for 6 hours. Total energy used?

d) Is there enough energy left to charge a phone (10 Wh)?

3 **Give two ways solar lights are better than kerosene lamps.** [2]

4 **Old batteries contain harmful materials. Suggest a way to deal with them safely.** [1]

Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

L	I	G	H	T	L	D	A	V	T	W
U	W	E	M	S	T	O	R	E	D	T
K	B	A	T	T	E	R	Y	N	F	K
C	I	I	N	S	U	L	A	T	O	R
O	S	N	N	H	J	M	C	N	F	B
I	O	E	E	C	D	K	Y	I	J	J
B	U	N	D	T	C	N	P	S	S	H
W	N	E	C	C	I	K	J	T	O	E
V	D	R	C	A	S	C	H	Y	L	E
H	G	G	H	E	A	T	C	K	A	R
V	G	Y	D	R	C	P	Y	J	R	H

battery **energy** **heat** **insulator** **kinetic** **light** **solar** **sound** **stored**

2 **Which form of energy is stored in food?** [1]

- ◯ kinetic
- ◯ chemical
- ◯ sound
- ◯ light

3 **A hot iron is left on a cold table. Which is true?** [1]

- ◯ the iron gains heat
- ◯ heat moves from the iron to the table
- ◯ cold moves into the iron
- ◯ the temperatures stay the same

4 **Draw an energy chain for a person playing a dhol (drum).** [2]

My energy chain

5 **Challenge. "We should save energy, but energy cannot be destroyed." Explain why both statements can be true.** [3]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name forms of energy with examples	○	○	○
draw energy chains	○	○	○
explain that energy is conserved but can be wasted	○	○	○
explain how temperature changes when heat is gained or lost	○	○	○
test which materials are good insulators	○	○	○
explain how a solar home system works	○	○	○